Exploring Convolutional Neural Processes for Weather Downscaling

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Abstract

This project is an exploration into using Convolutional Neural Processes (ConvNPs)for statistical downscaling of weather variables. The goal is to see how well these models work for this task. We'll use open data—coarse ERA5 grids and high-resolution GEBCO elevation—to predict weather at specific station locations. We will start with 2m temperature and then move to 10m wind fields. The core of the project is to get hands-on with ConvNPs, from data prep to training and evaluation.

Project description

The idea Global weather models give us the big picture, but their data is too coarse for local applications. We need a way to zoom in. This project will test if an uncertainty-aware deep learning method, Convolutional Neural Processes (ConvNPs) [1], is a good tool for the job.

ConvNPs are great for spatial data. They can take a few context points (like coarse weather data and elevation) and make predictions at new target locations (our weather stations). The plan is to see how effectively we can teach a ConvNP to learn the relationship between the large-scale weather patterns and the local conditions on the ground [2]. This is less about building a perfect downscaling system and more about exploring the methodology.

Project plan The approach is straightforward and broken down into a few key tasks:

1. **Understanding ConvNPs:** Spend time understanding what NPs [3] and ConvNPs [1] are, how they work, and why they might be a good fit for this problem. This means reading a few key papers and getting familiar with the concepts.
2. **Build the Dataset:**
   - Grab coarse temperature and wind data from the ERA5 dataset.
   - Get high-resolution elevation data from swisstopo
   - Download ground-truth station data from MeteoSwiss.
   - Combine everything into a unified dataset ready for training, keeping some measurement stations for validation.
3. **Train and Evaluate:** We'll use an existing codebase to hit the ground running.
   - Start with the simpler case: downscaling temperature.

- Once that's working, move on to the more complex wind fields.
- We'll test our model's performance using standard metrics and compare it to interpolated ERA5 data.

Expected outcome The main goal is to gain practical experience and assess the method. By the end, we should have:

- A clean dataset for training and evaluation
- A working ConvNP model trained to downscale temperature and wind.
- A clear quantitative answer to the question: "How well do ConvNPs work for this specific downscaling task?"
- A good understanding of the entire workflow, from raw data to model evaluation.
- Valuable hands-on experience with a state-of-the-art, uncertainty-aware deep learning model.

Additional information

- **Difficulty of the project:** The project is purely explorative, in terms of data and methodology. While the plan above is quite clear, it can be bent and adjusted anytime.
- **Requirements:** Machine Learning fundamentals, basic math skills, good Python skills, experience with git and pytorch
- **Nice-to-haves**: understanding of GPs and NPs
- **Supervisors:** Dr. Christian Donner (christian.donner@sdsc.ethz.ch)

References

[1] Jonathan Gordon, Wessel P Bruinsma, Andrew YK Foong, James Requeima, Yann Dubois, and Richard E Turner. Convolutional conditional neural processes. *arXiv preprint arXiv:1910.13556*, 2019.

[2] A. Vaughan, W. Tebbutt, J. S. Hosking, and R. E. Turner. Convolutional conditional neural processes for local climate downscaling. *Geoscientific Model Development*, 15(1):251–268, 2022.

[3] Marta Garnelo, Jonathan Schwarz, Dan Rosenbaum, Fabio Viola, Danilo J Rezende, SM Eslami, and Yee Whye Teh. Neural processes. *arXiv preprint arXiv:1807.01622*, 2018.